

# HACK: Supplementary Material

Additional accuracy results for “HACK: Homomorphic Acceleration via Compression of the Key-Value Cache for Disaggregated LLM Inference” (ACM SIGOPS ATC 2026)

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This document supplements the paper “HACK: Homomorphic Acceleration via Compression of the Key-Value Cache for Disaggregated LLM Inference” (ACM SIGOPS ATC 2026, <https://doi.org/10.1145/3828523.3857108>) with the per-dataset accuracy tables summarized in its Section 6.3. The code of HACK is at <https://github.com/Zeyu-ZEYU/HACK> (archived at <https://doi.org/10.5281/zenodo.22982679>). Table numbers below are local to this document.

## 1 Additional Accuracy Results

Tables 1–6 report per-dataset accuracy across all evaluated models; Table 7 reports additional metrics (perplexity, ROUGE-L, and pass@1/pass@10) for Qwen3 32B.

**Table 1.** Accuracy (IMDb).

|                    | Mis    | Phi    | Qwen   | Yi     | Lla    | Fal    |
|--------------------|--------|--------|--------|--------|--------|--------|
| Baseline           | 84.81% | 87.84% | 93.15% | 93.87% | 95.73% | 85.63% |
| HACK ( $\Pi=32$ )  | 83.79% | 87.08% | 93.02% | 92.82% | 94.73% | 84.96% |
| HACK ( $\Pi=64$ )  | 83.46% | 86.64% | 92.48% | 92.31% | 94.17% | 84.72% |
| CacheGen           | 83.04% | 86.15% | 91.19% | 91.79% | 93.85% | 83.94% |
| KVQuant            | 82.97% | 86.05% | 91.10% | 91.54% | 93.89% | 84.12% |
| HACK ( $\Pi=128$ ) | 82.71% | 86.07% | 90.75% | 91.35% | 93.76% | 84.03% |

**Table 2.** Accuracy (arXiv).

|                    | Mis    | Phi    | Qwen   | Yi     | Lla    | Fal    |
|--------------------|--------|--------|--------|--------|--------|--------|
| Baseline           | 79.40% | 86.35% | 88.50% | 87.75% | 83.79% | 79.42% |
| HACK ( $\Pi=32$ )  | 78.53% | 85.51% | 87.73% | 86.69% | 82.63% | 78.75% |
| HACK ( $\Pi=64$ )  | 78.26% | 85.08% | 87.25% | 86.39% | 82.31% | 78.28% |
| CacheGen           | 77.95% | 84.02% | 85.74% | 85.14% | 81.85% | 77.74% |
| KVQuant            | 77.64% | 84.11% | 85.50% | 85.08% | 81.76% | 77.72% |
| HACK ( $\Pi=128$ ) | 77.72% | 84.49% | 86.92% | 86.15% | 81.97% | 77.94% |

**Table 3.** Accuracy (Cocktail).

|                    | Mis    | Phi    | Qwen   | Yi     | Lla    |
|--------------------|--------|--------|--------|--------|--------|
| Baseline           | 75.18% | 83.92% | 89.25% | 85.25% | 86.39% |
| HACK ( $\Pi=32$ )  | 74.63% | 83.05% | 88.47% | 84.39% | 85.54% |
| HACK ( $\Pi=64$ )  | 74.42% | 82.66% | 88.06% | 84.04% | 84.87% |
| CacheGen           | 73.74% | 82.24% | 87.61% | 83.51% | 84.71% |
| KVQuant            | 73.72% | 82.26% | 87.58% | 83.42% | 84.68% |
| HACK ( $\Pi=128$ ) | 73.81% | 82.19% | 87.44% | 82.86% | 84.65% |

**Table 4.** Accuracy (HumanEval).

|                    | Mis    | Phi    | Qwen   | Yi     | Lla    | Fal    |
|--------------------|--------|--------|--------|--------|--------|--------|
| Baseline           | 89.37% | 91.62% | 92.17% | 90.79% | 92.45% | 85.21% |
| HACK ( $\Pi=32$ )  | 88.68% | 90.50% | 91.23% | 90.02% | 91.28% | 84.26% |
| HACK ( $\Pi=64$ )  | 88.19% | 90.16% | 90.85% | 89.63% | 91.04% | 83.82% |
| CacheGen           | 87.73% | 89.75% | 90.36% | 88.94% | 90.49% | 83.47% |
| KVQuant            | 87.64% | 89.65% | 90.27% | 88.93% | 90.43% | 83.29% |
| HACK ( $\Pi=128$ ) | 87.60% | 89.54% | 90.08% | 88.91% | 89.77% | 83.19% |

**Table 5.** Accuracy (GSM8K).

|                    | Mis   | Phi   | Qwen  | Yi    | Lla   | Fal   |
|--------------------|-------|-------|-------|-------|-------|-------|
| Baseline           | 96.1% | 94.3% | 95.9% | 96.2% | 95.6% | 83.7% |
| HACK ( $\Pi=32$ )  | 95.4% | 93.6% | 95.0% | 95.2% | 94.5% | 82.8% |
| HACK ( $\Pi=64$ )  | 95.2% | 93.2% | 94.6% | 94.7% | 94.1% | 82.3% |
| CacheGen           | 94.6% | 92.8% | 94.2% | 94.4% | 93.7% | 81.7% |
| KVQuant            | 94.5% | 92.7% | 94.1% | 94.4% | 93.5% | 81.6% |
| HACK ( $\Pi=128$ ) | 94.2% | 92.3% | 93.6% | 93.8% | 93.1% | 81.2% |

**Table 6.** Accuracy (RULER-128K).

|                    | Mis   | Phi   | Qwen  | Yi    | Lla   |
|--------------------|-------|-------|-------|-------|-------|
| Baseline           | 48.1% | 55.7% | 85.6% | 77.3% | 66.6% |
| HACK ( $\Pi=32$ )  | 47.7% | 55.2% | 85.1% | 76.9% | 66.2% |
| HACK ( $\Pi=64$ )  | 46.9% | 54.7% | 84.5% | 76.0% | 65.4% |
| CacheGen           | 46.0% | 53.9% | 83.7% | 75.4% | 64.8% |
| KVQuant            | 46.2% | 53.6% | 83.2% | 75.5% | 64.3% |
| HACK ( $\Pi=128$ ) | 45.1% | 52.8% | 82.4% | 74.2% | 63.6% |

**Additional metrics.** Beyond exact match, ROUGE-1, and edit similarity, Table 7 reports perplexity (PPL, WikiText-2 [1]), ROUGE-L (arXiv summarization), and pass@1/pass@10 (HumanEval) for Qwen3 32B. HACK ( $\Pi=32$ ) and HACK ( $\Pi=64$ ) track the BF16 baseline most closely and beat CacheGen and KVQuant on all four metrics, with the smaller partition slightly ahead, consistent with the exact-match results above.

**Table 7.** Additional accuracy metrics for Qwen3 32B.

|                   | WikiText-2 | arXiv   | HumanEval |         |
|-------------------|------------|---------|-----------|---------|
| Method            | PPL ↓      | ROUGE-L | pass@1    | pass@10 |
| Baseline          | 7.6        | 81.0%   | 87.0%     | 93.0%   |
| CacheGen          | 8.2        | 78.0%   | 83.5%     | 90.5%   |
| KVQuant           | 8.3        | 77.8%   | 83.0%     | 90.2%   |
| HACK ( $\Pi=32$ ) | 7.8        | 80.2%   | 86.0%     | 92.4%   |
| HACK ( $\Pi=64$ ) | 7.9        | 79.6%   | 85.3%     | 91.9%   |

## References

- [1] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2017. Pointer Sentinel Mixture Models. In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/1609.07843>